

Maya Elmosalamy

AI/ML Engineer

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Professional Summary

Recent Computer Science graduate with Honors (GPA 3.55/4.00) focused on AI/ML engineering, with hands-on projects in healthcare, NLP, computer vision, and autonomous robotics using Python, PyTorch, Scikit-learn, and Transformers.

Education

B.S. Computer Science, Misr International University
GPA: 3.55/4.00 | Graduated with Honors

Sep 2022 – Sep 2026

Experience

Summer Academy Intern, Dell Technologies

Aug 2025

- Completed 160+ hours of industry training covering Agile, software engineering practices, AI applications, and Dell Technologies' business and technology areas.
- Developed an outline for a proposed AI-driven approach to misinformation spread during crisis situations, considering business and technical requirements.

Projects

GeneX – AI-Driven Healthcare Decision Support System

Graduation Project

- Designed an AI-driven framework for autoimmune disease risk assessment and personalized drug-response analysis using genomic and clinical data.
- Preprocessed high-dimensional gene-expression data, performed feature engineering and dimensionality reduction, evaluated ML models for risk prediction, and integrated SHAP with drug-gene interaction analysis and digital-twin concepts.

Multilingual Fake News Detection – Arabic & English

- Built an Arabic-English text classification system for misinformation detection using multilingual text preprocessing, TF-IDF, and word embeddings.
- Trained and compared BiLSTM, CNN, and BERT models for text classification, achieving 93% accuracy on the evaluated dataset.

Autonomous Navigation Agent for Mobile Robotics

- Developed a perception-to-action pipeline that processed ultrasonic sensor data, constructed occupancy maps, detected obstacles, and planned collision-free paths using A* search.
- Integrated Python and Arduino through Bluetooth for live sensor telemetry and real-time navigation commands, with tools for path and movement visualization.

Lane Detection System

- Implemented a computer vision pipeline using grayscale conversion, Gaussian blur, Canny edge detection, and a region-of-interest mask for road-frame processing.
- Applied the Probabilistic Hough Line Transform to detect lane-line segments and overlaid detections onto video frames using OpenCV.

Credit Risk Assessment AI

- Built a machine learning pipeline for credit-status classification, including data merging, missing-value handling, categorical encoding, scaling, outlier filtering, and exploratory analysis.
- Applied a Genetic Algorithm for feature selection and evaluated KNN, Decision Tree, and MLP classifiers using Grid-SearchCV with separate training, validation, and test sets.

Technical Skills

Programming: Python, C++, Java, JavaScript, C# **ML/AI:** Scikit-learn, PyTorch, TensorFlow, XGBoost, Deep Learning, Model Evaluation, Feature Engineering, PCA, SHAP

NLP: Transformers, BERT, Multilingual NLP, Arabic NLP, Text Classification, TF-IDF, Word Embeddings **CV/Robotics:** OpenCV, Image Processing, Autonomous Agents, A* Search

Software/Data: Git, GitHub, Django, REST APIs, MySQL, NumPy, Matplotlib

Certifications

Artificial Intelligence Fundamentals Program – IBM SkillsBuild | Network Security Course – Cisco | Python Programming Basics – Mahara-Tech

Languages

Arabic: Native | English: B2 | German: B1